

$$\begin{cases} A \text{ (защита)} & - & x & \text{у.е.} \\ B \text{ (атака)} & - & y & \text{у.е.} \end{cases}$$

①

Кол-во инф НСД: $J(x, y)$

$$\begin{cases} f(z) & - \text{затраты на созд. и хран. } z \text{ у.е.} \\ g(z) & - \text{ценность } z \text{ у.е. для } B \end{cases}$$

Затраты А: $u(x, y) = f[J(x, y)] + x \rightarrow \min_x$

Прибыль В: $v(x, y) = g[J(x, y)] - y \rightarrow \max_y$

ОПТИМАЛЬНЫЕ стратегии
(ИЭШ)

II

$$\begin{cases} \frac{\partial u}{\partial x} = f'[I(x, y)] \frac{\partial I(x, y)}{\partial x} + 1 = 0 \\ \frac{\partial v}{\partial y} = g'[J(x, y)] \frac{\partial J(x, y)}{\partial y} - 1 = 0 \end{cases}$$

$$\begin{cases} x^* = F(J, f, g) \\ y^* = G(J, f, g) \end{cases}$$

Пример

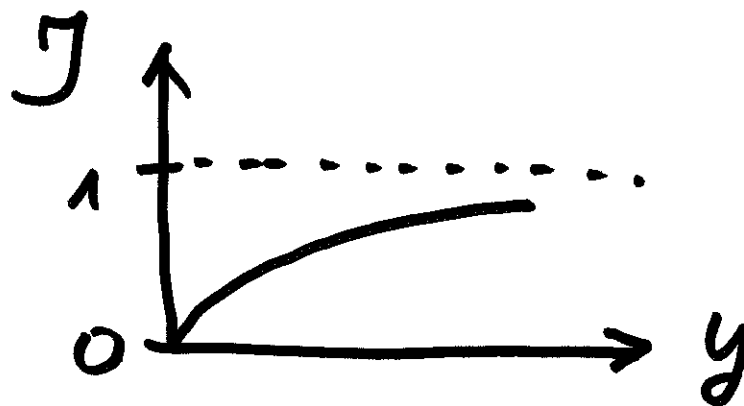
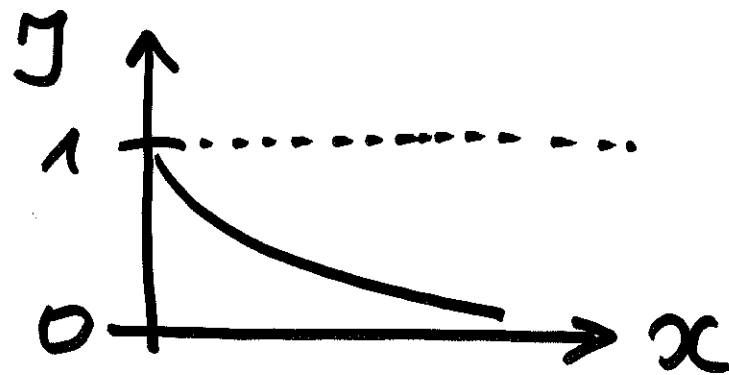
III

$$\boxed{J(x, y) = e^{-x} (1 - e^{-y})}$$

$$\begin{cases} J'_x = (e^{-y} - 1)e^{-x} = -J(x, y) \\ J'_y = e^{-(x, y)} \end{cases}$$

$$\begin{cases} f(z) = az, & a > 0 \\ g(z) = bz, & b > 0 \end{cases}$$

$$\begin{cases} f'(z) = a \\ g'(z) = b \end{cases}$$



$$\begin{cases} \frac{\partial u}{\partial x} = -ae^{-x}(1-e^{-y})+1=0 \\ \frac{\partial v}{\partial y} = be^{-(x+y)}-1=0 \end{cases}$$

IV

$$\begin{cases} x^* = \ln \frac{ab}{a+b} \\ y^* = \ln \frac{a+b}{a} \end{cases}$$